Guided Portfolio Watch — Design Process

Problem and Initial Persona

With limited user research available for this assignment, I started with a problem I knew from personal experience: most financial dashboards assume the reader already understands finance. My initial persona hypothesis was a first-time long-term investor — someone who bought a few equities, intends to hold 3–5 years, and has no financial background. This was a working assumption tested through limited early usability sessions, not a market-validated persona.

The problem worth solving: *show only what materially affects a long-term business case, in plain English, without requiring prior financial knowledge.*

User Insight and Product Hypothesis

The core insight came from early usability testing. A prototype displaying accurate financial metrics — "revenue up 65%, net income up 64%" — failed to communicate anything meaningful to a non-expert tester. The numbers were correct; the context was absent.

This surfaced the hypothesis: **the right unit of information is not a metric; it is an explanation.** Every alert and insight card leads with *what happened*, follows with *why it matters*, and closes with *what to watch next* — with sources. In guided mode, metric labels read "Total sales," "Profit after costs," "Cash generated."

Major Design Decisions

No price data. A beginner monitoring prices daily is likely to over-react and undermine a long-term strategy. The Skill covers material business-level changes only: earnings, regulatory decisions, strategic announcements, supply-chain shifts.

Guidance inferred, not asked — with explicit decision boundaries. Asking "are you a beginner?" is a friction point. Guidance mode is inferred from natural-language signals, overridable via AGENTS.md. Prompt instructions in `buildPreferencesNote()` direct the LLM to surface evidence, uncertainties, and questions rather than recommend buying, selling, or sizing positions. Alert formatting reinforces this; actual LLM output compliance is unverified.

Evidence gate on alerts. After the first production run, the selected finding had no external source URL. Alerts without at least one verifiable HTTPS citation are suppressed; source URL coverage is an open product question.

Single config as source of truth. Both feeds read from one `config.json` at runtime. Changing tickers produces a fully updated Playbook and alert pipeline with no code changes required.

Skill Workflow and Testing

Setup is a 7-step agent workflow. `parseIntent()` maps input deterministically to a `MonitoringConfig` — no cloud calls until the user confirms.

404 tests cover intent parsing, YoY logic, Playbook rendering, reusability, alert pipeline, and the decision-support contract. 93 monitor-alert tests were added after a production run surfaced data-contract bugs and the source URL gap. One production run completed (110 events, 1 finding selected); alert generation failed due to a post-processing bug and was fixed. No end-to-end financial-event notification has been delivered.